

Time: 20 mints

## SECTION-A (Marks 15)

Total Marks:75

| Question:1                                                                           | A                                                                                   | B                                                                                   | C                                                                                    | D                                                                                     | A                     | B                     | C                     | D                     |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 1.if $\cos x = \sin x$ , then the value of x is :                                    | $30^0$                                                                              | $45^0$                                                                              | $90^0$                                                                               | $60^0$                                                                                | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 2.if $\sin x = \frac{1}{4}$ what is the value of value of $\cos x$                   | $\frac{3}{4}$                                                                       | $\frac{\sqrt{15}}{4}$                                                               | $\frac{\sqrt{17}}{4}$                                                                | $\frac{15}{4}$                                                                        | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 3.which one is the linear equations in one variable :                                | $ax + y = 0$                                                                        | $yx + 3 = 0$                                                                        | $2x + 3 = 0$                                                                         | $3x^2 + 2 = 0$                                                                        | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 4. Different Number of ways to describe the set are :                                | 1                                                                                   | 2                                                                                   | 3                                                                                    | None                                                                                  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 5. $\frac{2x+1}{(x+1)(x-1)}$ is a _____                                              | Equation                                                                            | Proper fraction                                                                     | Improper fraction                                                                    | None                                                                                  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 6. The Measure which determines the middle most observation in a data set ,is called | Mode                                                                                | Harmonic                                                                            | Arithmetic Means                                                                     | Median                                                                                | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 7. $\frac{3\pi}{4}$ radian = _____                                                   | $115^0$                                                                             | $135^0$                                                                             | $150^0$                                                                              | $30^0$                                                                                | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 8. $\sec \theta \cot \theta =$                                                       | $\sin \theta$                                                                       | $\frac{\sin \theta}{\cos \theta}$                                                   | $\frac{1}{\cos \theta}$                                                              | $\frac{1}{\sin \theta}$                                                               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 9. The symbol for rectangular is denoted by:                                         |  |  |  |  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 10. If $x < 0, y > 0$ , then $(-x, y)$ will lie in quadratic                         | I                                                                                   | II                                                                                  | III                                                                                  | IV                                                                                    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 11. Minimum number of sides in a polygon is :                                        | 2                                                                                   | 3                                                                                   | 5                                                                                    | 4                                                                                     | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 12. The line $5x - ky - 3 = 0$ passes through (1,2).What is K                        | -2                                                                                  | -1                                                                                  | 1                                                                                    | 2                                                                                     | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 13. Slope of two lines are 0 and $\infty$ . What is the angle between both lines ?   | $0^0$                                                                               | $45^0$                                                                              | $60^0$                                                                               | $90^0$                                                                                | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 14. Which of the following is polygon ?                                              | Circle                                                                              | Pyramid                                                                             | Quadrilateral                                                                        | Sphere                                                                                | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 15. How many sides of a regular polygon has if its exterior angle is $15^0$ ?        | 20                                                                                  | 21                                                                                  | 24                                                                                   | 25                                                                                    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

| Q2     | Question                                                                                                                                                                                                             | Marks |    | Question                                                                                                                                                                                                                                                                                                  | Marks |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| (i)    | Use properties of exponents to simplify<br>$\sqrt[4]{\frac{a^3}{b^3}} \times \sqrt[4]{\frac{b^3}{c^3}} \times \sqrt[4]{\frac{c^3}{a^3}}$                                                                             | 4     | OR | Simplify<br>$\left(\frac{2187 a^5 b^{17}}{a^{12} b^3}\right)^{\frac{1}{7}}$                                                                                                                                                                                                                               | 4     |
| (ii)   | The Kansu ,China earthquake of 1920 was measured about 8.5 on Richter scale and the Tokyo ,Japan earth of 1923 was measured 7.8 on that scale how many times stronger was the 1920 earthquake than 1923 earthquake ? | 4     | OR | If $\log b^2 = 0.3010$ , $\log b^3 = 0.4771$ $\log b^5 = 0.6990$ then evaluate the following with laws of logarithm<br>(i) $\log b^{0.024}$                                                                                                                                                               | 4     |
| (iii)  | If $X = \{a, b, c, d, e\}$ , $Y = \{a, c, e\}$ , $Z = \{g, h, i, j\}$ then find the following using Venn diagram<br>(XUY)UZ                                                                                          | 4     | OR | If $L = \{a, b, c\}$ $M = \{d, e, f, g\}$ then find two binary relation in $L \times M$                                                                                                                                                                                                                   | 4     |
| (iv)   | Find the HCF of $18ab^3$ , $30a^2b^4c^3$ , $24b^2c^5$                                                                                                                                                                | 4     | OR | Find LCM of expression :<br>$p^3q - pq^3, p^5q^2 - p^2q^5$                                                                                                                                                                                                                                                | 4     |
| (v)    | Solve $\frac{4(x+2)}{3} - \frac{6(x-7)}{7} = 12$                                                                                                                                                                     | 4     | OR | Solve the following inequalities<br>$16 - 5y < 4(y - 1) - 7, y \in R$                                                                                                                                                                                                                                     | 4     |
| (vi)   | Convert the following into degree<br>$45^\circ 45' 45''$                                                                                                                                                             | 4     | OR | Find $r$ when $l = 52cm$ , $\theta = 45^\circ$                                                                                                                                                                                                                                                            | 4     |
| (viii) | Find Arithmetic mean by direct method for following set of data: 200,225,350,375,270,320,290                                                                                                                         | 4     | OR | Find garment factory shipped an order which contained 7000 jackets ,2000 sweaters, and 3000 trousers in 7 trousers ,5 jackets and 9 sweaters are faulty. If the quality inspector unpacked one items at random, find the probability that it is<br>(i) a trouser (ii) not a jacket<br>(iii) a faulty item | 4     |
| (ix)   | Find gradient and inclination of lines joining:<br>$A(2,6), B(5,8)$                                                                                                                                                  | 4     | OR | Reduce the given equations into slope -intercept form<br>$6x + 8y - 11 = 0$                                                                                                                                                                                                                               | 4     |

1. Find HCF of the following by division method

$$2x^3 + 6x^2 + x + 3, 3x^3 + 9x^2 - 2x - 6, x^3 + 3x^2 + 2x + 6$$

2. Volume of a cubical oil tank in an oil refinery is expressed as  $(125x^3 - 150x^2 + 60x - 8)m^3$ . Find; a) expression for height of oil tank. b) expression for surface area of oil tank c) expression for painting it from outside @Rs32/ $m^2$

3. Find the square roots of  $(a + \frac{1}{a})^2 - 4(a - \frac{1}{a})$ .

4. Find square root of  $4(x^2 + \frac{1}{x^2}) + 12(x + \frac{1}{x}) + 17$ .

5. Find the second polynomial when

first polynomial =  $x^4 + x^3 + x + 1$ , HCF =  $x + 1$  and

LCM =  $(x^3 + 1)(x^4 + x^3 - x - 1)$

6. For what value of k, the expression

$y^4 + 4y^2 + k + \frac{8}{y^2} + \frac{4}{y^4}$  becomes a perfect square, where  $y \neq 0$ .

7. Find the HCF of the following by factorization

$$ax^2 + 2a^2 + a^3, 2ax^2 - 4a^2x - 6a^3, 3(ax + a^2)^2$$

8. Find the HCF of the following by factorization

$$m^2 - n^2, m^4 - n^4, m^6 - n^6$$

9. Factorize the expression completely

$$(5 - x)(2 + x)(10 - x)(1 + x) - 7x^2$$

10. Factorize the expression completely

$$(2x^2 - 3x + 6)(2x^2 - 3x) - 55$$

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| Chapter 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p>1. If <math>\sin\theta = \frac{\sqrt{3}}{2}</math> and terminal ray of <math>\theta</math> is in first quadrant, then find the remaining trigonometric ratios for same value of <math>\theta</math>.</p> <p>2. Solve the right angled triangle if <math>\triangle ABC</math>, <math>\angle C = 90^\circ</math>, <math>a=4\text{cm}</math>, <math>c=4\sqrt{2}\text{cm}</math></p> <p>3. from A point at distance of 20 m from a tree, angle of elevation of a tree is <math>30^\circ</math>. find height of the tree</p> <p>4. An Aeroplane at an altitude of 900 m find that two ships are sailing towards it in the same direction. The angles of depression of ships as observed from plane are <math>30^\circ</math> and <math>60^\circ</math>. Find distance between ships.</p> <p>5. Prove that <math>\sqrt{\frac{1-\cos\theta}{1+\cos\theta}} = \frac{\sin\theta}{1+\cos\theta}</math></p> <p>6. Show that <math>\sec^2\theta - \tan^2\theta = 1</math></p> <p>7. Show that <math>\sin^2\theta - \cos^2\theta = 1</math></p> <p>8. A Ship sails from port A to Port B at a bearing of <math>040^\circ</math> for 35 km. Then, it sails at a bearing of <math>135^\circ</math> for 50 km from B to port C. Find</p> <p>(a) The distance between port A and port C</p> <p>(b) The bearing of C from A</p> <p>9. The positions of three ships A, B and C are such that the bearing of B from A is <math>045^\circ</math> and ship C is to the east of A. if bearing of C from B is <math>180^\circ</math> and <math>mAB = 10\text{km}</math>, what is the bearing of A from C? Also find the distance between ship A and ship C.</p> |

1. Find the equations of family of lines passing through point of intersection of  $6x + 5y + 3$  and  $2x - 5y + 3 = 0$  and slope 3.
2. Find the interior angles of triangles whose vertices are A(-2,0) B(3,0) and C(6,5)
3. Reduced the equation  $6x-5y+15=0$ 
  - (1) Slope intercept form
  - (2) two intercept form
  - (3) Point slope form
  - (4) Two point form
  - (5) Normal form
  - (6) Symmetric form
4. Find the equation of straight line
  - (1) Through (3,-2) and perpendicular to the line with slope 3
  - (2) Having y intercept = -7 and parallel to the line with slope 4
  - (3) Through (-5,1) and perpendicular to the line passing through (0,9) and (-2,6)
  - (4) If the length of perpendicular from the origin is 6 units inclined at  $60^\circ$

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| Chapter 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p>1. Construct a <math>\triangle LMN</math> when <math>LM=4.1</math> cm ,<math>MN=4</math> cm and <math>\angle M = 90^0</math>. Draw perpendicular bisectors of sides of triangles .Are they concurrent .</p> <p>2. Construct a <math>\triangle XYZ</math> when <math>XY=6</math> cm and <math>\angle X = 45^0</math> And <math>\angle Y = 60^0</math>. Draw altitudes of triangles .Are they concurrent .</p> <p>3. Construct a <math>\triangle ABC</math> when <math>AB=4.7</math> cm ,<math>BC=4.5</math> cm and <math>AC=4.6</math>cm . Draw bisectors of Three angle of triangles .Are they concurrent</p> <p>4. Construct a triangle and prove that the median of a triangles is concurrent .(example 9)</p> <p>5. Show that the median of the following triangle is concurrent if <math>\triangle ABC</math> when <math>AB=5.8</math> cm ,<math>BC=5</math> cm and <math>\angle Y = 45^0</math>.</p> |

## Chapter 11

- 1. Find mean ,median and mode for the following discrete frequency distribution**

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| x | 0 | 1 | 2 | 3 | 4 | 5 |
| f | 3 | 5 | 9 | 4 | 2 | 1 |

- 2 Find modal value for the following frequency distribution**

|         |     |     |      |       |       |
|---------|-----|-----|------|-------|-------|
| classes | 1-4 | 5-8 | 9-12 | 13-16 | 17-20 |
| f       | 9   | 8   | 10   | 6     | 7     |

- 3 Marks obtained by 100 students in an examinations are given below**

|               |       |       |       |       |       |
|---------------|-------|-------|-------|-------|-------|
| Marks         | 10-19 | 20-29 | 30-39 | 40-49 | 50-59 |
| No of student | 5     | 25    | 40    | 20    | 10    |